Homework 1

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Problem 1

By definition, $\varnothing, \mathcal{T} \in \mathbb{R}$. Now let $\{(-\infty, p)\}_{p\in P} \subseteq \mathcal{T}$, and consider the union over this collection. Let $a = \sup P$, and let $x \in (-\infty, a)$. We have that $x < \sup P$, and thus by definition, there exists $p \in P$ such that $x < p$. It follows that $x \in (-\infty, p)$, and thus we have

$$x \in (-\infty, p) \subseteq \bigcup_{p\in P} (-\infty, p).$$

Therefore, $(-\infty, a) \subseteq \bigcup_{p\in P}(-\infty, p)$. Now let $x \in \bigcup_{p\in P}(-\infty, p)$. It follows that there exists $p \in P$ such that $x \in (-\infty, p)$, and thus $x < p$ for some $p \in P$. Since $p \leq \sup P = a$, we have that $x < a$, and thus $x \in (-\infty, a)$. Therefore,

$$\bigcup_{p\in P} (-\infty, p) = (-\infty, \sup P).$$

Since $(-\infty, \sup P)$ is of the same form as given in the definition of $\mathcal{T}$, we have $(-\infty, \sup P) \in \mathcal{T}$. Therefore, $\mathcal{T}$ is closed under union.

Now let P be a finite subset of $\mathbb{R}$, and consider the collection $\{(-\infty, p)\}_{p\in P}$. Since P is finite, it must have a least element. Let $a = \min P$ be this least element, and note that for all $p \in P$, $a \leq p$ implies that $(-\infty, a) \subseteq (-\infty, p)$. It follows that

$$(-\infty, a) \subseteq \bigcap_{p\in P} (-\infty, p).$$

However, we also know that since $(-\infty, a)$ is one of the elements the intersection is over,

$$\bigcap_{p\in P} (-\infty, p) \subseteq (-\infty, a).$$

Therefore, the finite intersection is of the form $(-\infty, a)$, and thus is an element of $\mathcal{T}$. We have that $\mathcal{T}$ is closed under finite intersection, and is thus a topology on $\mathbb{R}$.

Problem 2

Let $n \in \mathbb{Z}$. If n is odd, we have $n \in \{n\} = B(n) \in \mathcal{B}$. If n is even, then $n \in \{n-1, n, n+1\} = B(n) \in \mathcal{B}$. Therefore, for all $n \in \mathbb{Z}$, there exists $B(n) \in \mathcal{B}$ such that $n \in B(n)$.

Now consider the intersection $B(n) \cap B(m)$ for $n, m \in \mathbb{Z}$. If $B(n) \cap B(m) = \varnothing$, then it is vacuously true that for all $x \in B(n) \cap B(m)$, there exists $B \in \mathcal{B}$ such that $x \in \mathcal{B} \subseteq B(n) \cap B(m)$. Thus, assume without loss of generality that the intersection is nonempty.

Consider the case where either n or m is odd. Without loss of generality, assume n is odd. We thus have

$$n \in B(n) \cap B(m) \implies B(n) = \{n\} \subseteq B(n) \cap B(m).$$

Now consider the case where both n and m are even. If $n = m$, we trivially have $B(n) \subseteq B(n) \cap B(m)$ once again, since $B(n) = B(m)$, so assume without loss of generality that $n < m$. We then have

$$B(n) \cap B(m) = \{n-1, n, n+1\} \cap \{m-1, m, m+1\}\,.$$

Since $m > n$, but the intersection is nonempty, we must have $m - 1 = n + 1$, since if m is any greater the intersection is nonempty, but if m is any less than $m = n$ or $m < n$. Note that since n is even, $n + 1$ is odd, so

$$B(n+1) \subseteq B(n) \cap B(m).$$

In all of these cases, note that $B(n), B(n+1) \in \mathcal{B}$, so we have that $\mathcal{B}$ is a basis for a topology on $\mathbb{Z}$.

Problem 3

Note that $X^c = X \setminus X = \varnothing$, and $\varnothing^c = X \setminus \varnothing = X$. Therefore, $\varnothing, X$ are closed.

Let $\{X_a\}_{a \in A}$ be a collection of closed sets in X. It follows that there exists a collection $\{Y_a\}_{a \in A}$ of open sets such that $Y_a^c = X_a$. It follows that

$$\bigcap_{a \in A} X_a = \bigcap_{a \in A} Y_a^c.$$

By De Morgan's laws, this equals

$$\left(\bigcup_{a \in A} Y_a\right)^c.$$

By the definition of a topology, $\bigcup_{a \in A} Y_a$ is open, so $\left(\bigcup_{a \in A} Y_a\right)^c$ is closed.

Let $\{X_k\}_{i=1}^k$ be a finite collection of closed sets in X. It follows that there exists a collection $\{Y_i\}_{i=1}^k$ of open sets such that $Y_i^c = X_i$ for all i. By De Morgan's laws, it follows that

$$\bigcup_{i=1}^k X_i = \bigcup_{i=1}^k Y_i^c = \left(\bigcap_{i=1}^k Y_i\right)^c.$$

By definition of a topology, since Y_i is open for all i, it follows that $\bigcap_{i=1}^k Y_i$ is open. Therefore, $\left(\bigcap_{i=1}^k Y_i\right)^c$ is closed.

Problem 4

Let $(x, y) \in \mathbb{R}^2$, and choose some $\varepsilon \in \mathbb{R}_{>0}$. We have

$$(x, y) \in (x - \varepsilon, x + \varepsilon) \times (y - \varepsilon, y + \varepsilon) \in \mathcal{B}.$$

Now consider the intersection $A \cap B$, where

$$A \coloneqq (a_1, b_1) \times (c_1, d_1), \qquad B \coloneqq (a_2, b_2) \times (c_2, d_2).$$

Let $(x, y) \in A \cap B$. I claim

$$(x, y) \in (\max(a_1, a_2), \min(b_1, b_2)) \times (\max(c_1, c_2), \min(d_1, d_2)) \subseteq A \cap B.$$

First, note that if $(x, y) \in A \cap B$, then we must have

$$a_1 < x < b_1 \text{ and } a_2 < x < b_2.$$

Therefore, $\max(a_1, a_2) < x < \min(b_1, b_2)$. Similarly, we have

$$c_1 < y < d_1 \text{ and } c_2 < y < d_2,$$

and thus $\max(c_1, c_2) < y < \min(d_1, d_2)$. Therefore,

$$(x, y) \in (\max(a_1, a_2), \min(b_1, b_2)) \times (\max(c_1, c_2), \min(d_1, d_2)).$$

It remains to be shown this set is a subset of the intersection $A \cap B$. Let (x, y) in this set. It follows that $\max(a_1, a_2) < x < \min(b_1, b_2)$, and thus

$$\begin{aligned} a_1 &< x \\ a_2 &< x \\ b_1 &> x \\ b_2 &> x. \end{aligned}$$

Therefore, $x \in (a_1, b_1)$ and $x \in (a_2, b_2)$. Applying the same logic to c and d shows $y \in (c_1, d_1)$ and $y \in (c_2, d_2)$. Therefore, we have

$$(x, y) \in ((a_1, b_1) \times (c_1, d_1)) \cap ((a_2, b_2) \times (c_2, d_2)).$$

Therefore,

$$((\max(a_1, a_2), \min(b_1, b_2)) \times (\max(c_1, c_2), \min(d_1, d_2))) \subseteq A \cap B.$$

Therefore, $\mathcal{B}$ is a basis on $\mathbb{R}$. It remains to be shown $\mathcal{B}$ generates the standard topology.

Let $\mathcal{B}_S$ be the standard basis, $\mathcal{T}_\mathcal{B}$ the topology generated by $\mathcal{B}$, and $\mathcal{T}_S$ the topology generated by $\mathcal{B}_S$. It suffices to show $\mathcal{B}_S \subseteq \mathcal{T}_\mathcal{B}$ and $\mathcal{B} \subseteq \mathcal{T}_S$, since any topology containing one of these bases must also contain all sets generated by these bases. First, we show $\mathcal{B} \subseteq \mathcal{T}_S$.

Let $U := (a, b) \times (c, D) \in \mathcal{B}$. Let $(x, y) \in U$, and let $d_{(x,y)}$ be the *minimum* distance from (x, y) to the edge of U. That is,

$$d_{(x,y)} = \min \{d(x, a), d(x, b), d(y, c), d(y, D)\}.$$

For simplicity of notation, we will now refer to points with a single letter, rather than as a pair. Let $B(x, d_x)$ be the open ball of radius d_x as defined previously, centered at x. Note that if $x \in B(x, d_x)$, then it must be a shorter distance away from x than the distance to the edge of the rectangle U, so $x \in U$. Therefore, $B(x, d_x) \subseteq U$ for all $x \in U$. It follows that

$$\bigcup_{x \in U} B(x, d_x) \subseteq U,$$

since all sets $B(x, d_x) \subseteq U$. However, since this union contains all points $x \in U$, we must also have

$$U \subseteq \bigcup_{x \in U} B(x, d_x) \implies U = \bigcup_{x \in U} B(x, d_x).$$

Therefore, $U \in \mathcal{T}_S$, and $\mathcal{B} \subseteq \mathcal{T}_S$.

Now let $U \coloneqq B(k, r) \in \mathcal{T}_{\mathcal{B}}$. Consider a square $(a, b) \times (c, d)$ centered at (x, y); that is, $b - a = d - c$ and

$$x = \frac{b-a}{2}, \qquad y = \frac{d-c}{2}.$$

Note that the distance $d_{(x,y)}$ from (x, y) to a corner of the square would be, by the pythagorean theorem,

$$d_{(x,y)} = \left(\frac{b-a}{2}\right)^2 + \left(\frac{d-c}{2}\right)^2 = \frac{1}{4}\left(a^2 + b^2 + c^2 + d^2 - 2(ab + cd)\right).$$

Define A_x to be the open square centered at x, where $d_x = r$. Note that for all $a \in A_x$, we must have $d(x, a) < r$, since the square is open, so $A_x \subseteq B(k, r)$ provided that $x \in B(k, r)$. Consider the union

$$\bigcup_{x \in B(k,r)} A_x.$$

Note that since this union contains all $x \in B(k, r)$, we must have $B(k, r)$ as a subset of it. Additionally, since each set in this union is a subset of $B(k, r)$, the entire union must be a subset of $B(k, r)$. Therefore,

$$B(k, r) = \bigcup_{x \in B(k,r)} A_x.$$

Since $A_x \in \mathcal{B}$, this is a union over elements of the basis, and thus $B(k, r) \in \mathcal{T}_{\mathcal{B}}$. Therefore, $\mathcal{B}_S \subseteq \mathcal{T}_{\mathcal{B}}$.

Problem 5

Let X be Hausdorff, and consider the intersection

$$U \coloneqq \bigcap_V \overline{V},$$

where V ranges over all open sets in X containing x. By definition, we have $x \in \overline{V}$ for all V, so $x \in U$. Note that for all $y \in X$ such that $y \neq x$, since X is Hausdorff, there exist disjoint neighborhoods V_x, V_y of x, y, respectively. Suppose for contradiction that $y \in U$. Note that $\overline{V_x} \supseteq U$, and because $y \notin V_x$, we must have $y \in \overline{V_x} \setminus V_x = \partial V_x$, because V_x is open and thus $\operatorname{Int}(A) = A$. It follows that y is a limit point of V_x, and thus every open set containing U must intersect V_x at a point other than y. Therefore, V_y intersects V_x at some point other than y, and $V_y \cap V_x$ must thus contain this point, and is thus nonempty. This is a contradiction, and thus $y \notin U$.

Now for the converse, suppose $U = \{x\}$. Let $y \in X$ be a distinct point. Since $y \notin U$, it follows that there must exist at least one neighborhood V of x such that $y \notin \overline{V}$. Equivalently, $y \in \overline{V}^c$. Note that $\overline{V} \cap \overline{V}^c = \varnothing$. Since $V \subseteq \overline{V}$, we have $V \cap \overline{V}^c = \varnothing$. Since $\overline{V}$ is closed, $\overline{V}^c$ is open, and contains y. V is also open, and contains x. Therefore, for any two distinct points x, y, there exist disjoint neighborhoods $V, \overline{V}^c$ of those points, and X is Hausdorff.

Problem 6

Let definitions be as given in the problem, and suppose X is Hausdorff. Then for all $x \neq y$ in X, there exist disjoint neighborhoods V_x, V_y of x and y, respectively. Equivalently, $V_x \subseteq X \setminus V_y = V_y^c$. Since V_y is open, V_y^c is finite, and thus V_x is finite. Note that $X = V_x \cup V_x^c$. Since V_x and V_x^c are finite, X is finite.

Now suppose X is finite. It follows that all subsets $U \subseteq X$ are finite, and thus $U^c = X \setminus U$ is finite. Therefore, $\mathcal{T} = \mathscr{P}(X)$ is the power set of X, since every possible set has finite compliment, and we have that for any points $x \neq y$ in X, the sets $\{x\}, \{y\} \in \mathcal{T}$ are trivially disjoint.

Problem 7

Note that for any $C \in \mathcal{C}$, the interesection $\bigcap_{i=1}^{1} C = C \in \mathcal{B}_{\mathcal{C}}$. Therefore, $\mathcal{C} \subseteq \mathcal{B}_{\mathcal{C}}$. By definition,

$$\bigcup_{C \in \mathcal{C}} C = X.$$

Therefore, for all $x \in X$, there exists $C \in \mathcal{B}_{\mathcal{C}}$ such that $x \in C$. Now let $U_1, U_2 \in \mathcal{B}_{\mathcal{C}}$. It follows that there exist finite collections $\{C_i\}_{i=1}^{k}, \{C_j\}_{j=1}^{\ell} \subseteq \mathcal{C}$ such that

$$U_1 = \bigcap_{i=1}^{k} C_i, \qquad U_2 = \bigcap_{j=1}^{\ell} C_j.$$

Let $\mathcal{V}$ be the collection of all C_is and C_js. We have

$$U_1 \cap U_2 = \left(\bigcap_{i=1}^{k} C_i \right) \cap \left(\bigcap_{j=1}^{\ell} C_j \right) = \bigcap_{V \in \mathcal{V}} V.$$

Since $\{C_i\}_{i=1}^{k}$ and $\{C_j\}_{j=1}^{\ell}$ are finite, $\mathcal{V}$ is finite, and thus $U_1 \cap U_2$ is a finite intersection of elements of $\mathcal{C}$. Therefore, $U_1 \cap U_2 \in \mathcal{B}_{\mathcal{C}}$. Note that

$$U_1 \cap U_2 \subseteq U_1 \cap U_2,$$

and thus $\mathcal{B}_{\mathcal{C}}$ is a basis.